

A dynamic crowding-impact universe for 13F positioning

Quarterly 13F holdings from SEC EDGAR · 2015Q1–2024Q4 · config a8e53b995d2f

All performance figures in this memo come from generated fixtures, not from markets. The fixture generator writes EDGAR-shaped bytes so the parsing and point-in-time layers are genuinely exercised, and it plants a known price-pressure mechanism so the pipeline can be checked for its ability to recover a signal that is known to be there. It cannot tell you whether the factor works. Read the results section as evidence about the plumbing.

The problem with a fixed manager list

The standard construction of a 13F crowding factor starts from a hand-picked set of managers – the largest funds plus a few famous names. That choice carries three defects, and the third is the one that quietly does the damage.

- **Selection on survivors.** A list drawn up today and applied from 2015 embeds knowledge of who would still matter in 2025. The managers who blew up are missing, and their forced liquidations were precisely the crowding events the factor claims to measure.
- **Size is not footprint.** A \$50bn book held in megacaps can be unwound in days and moves nothing. A \$3bn book in illiquid small caps moves prices on the way out. Ranking managers by assets ranks the wrong quantity.
- **The normalisation trap.** Treat every listed manager as one equal voice and a single-position wrapper vehicle counts for as much as a diversified multi-strategy platform. The signal is then dominated by filers with no capacity to transmit anything.

So the universe is not chosen. It is *derived*, every quarter, from a rule that ranks managers on their lagged capacity to transmit common demand shocks into prices. Selection never touches realised performance: the inputs are mechanical properties of the book, observable at the time.

The selection rule

$\text{Score}(m, t) = 0.40 \cdot \text{rank}(\text{Impact}) + 0.40 \cdot \text{rank}(\text{Centrality}) + 0.20 \cdot \text{rank}(\text{Turnover})$, each a cross-sectional percentile rank within the quarter. The universe for quarter t is the top 25 by score at $t-1$.

- **Impact** – the liquidity footprint of the book: for each position, $\text{value} \times \text{daily volatility} \times (\text{value}/\text{ADV})^{0.5}$, summed. The square-root exponent is the standard impact law; it is a config parameter, not a constant buried in a function, and 1.0 gives the linear alternative.
- **Centrality** – liquidity-weighted portfolio overlap with every other filer. Computed as $\mathbf{x}_m \cdot \mathbf{s} - \|\mathbf{x}_m\|^2$ rather than as a pairwise loop, which is $O(M \cdot N)$ instead of $O(M^2 \cdot N)$ and is the difference between seconds and hours at real filer counts. A test asserts the shortcut equals the naive double loop.
- **Turnover** – half the sum of absolute split-adjusted share changes priced at a common date, EWMA-smoothed over 3 quarters. A manager who never trades cannot transmit a shock however large the book.

Eligibility precedes scoring: at least 4 quarters of history, a \$500mm equity book, 15 positions, and no single name above 60% of the book. That last filter is what removes the single-position wrapper.

Membership uses hysteresis: a manager enters at rank 25 or better and leaves only past rank 35. Without the band, names sitting near the threshold oscillate in and out on rank noise and the portfolio pays transaction costs for churn that carries no information.

Point-in-time discipline

Every 13F fact carries two dates, and conflating them is the standard way a 13F backtest acquires lookahead: *valid time*, the quarter the positions describe, and *transaction time*, the instant the fact became public. The store answers exactly one question – `as_of(period_end, knowledge_date)` – and the factor layer is not permitted to reach around it. Both the current and prior quarter are always materialised at the same vantage, so a quarter-on-quarter position change never mixes what was known then with what is known now.

The deadline is a floor, not a schedule

Measured over 40 quarters: median lag 41 days, p90 59, worst 159. Only 83% of filers are in by day 45 in the average quarter, and 77% in the worst. Trading on day 46 means trading without roughly one filer in six.

So the rebalance calendar is derived from the data, not the regulation. A quarter activates on the first rebalance date where 80% of *the selected universe* has actually filed, with a hard backstop at 150 days so one chronically late filer cannot stall the period indefinitely – silently dropping quarters would be a survivorship filter on periods. A further business day separates readable from tradeable.

Realised activation lag: median 61 days, p90 92, max 123.

Amendments and confidential treatment

Amendment semantics follow the cover page rather than a guess. **RESTATEMENT** replaces the table for that filer-quarter. **NEW HOLDINGS** is additive and carries only rows omitted from the original – the shape a lapsed confidential-treatment order takes, where the positions genuinely were not in the public record at the original date. An unflagged amendment is treated as a restatement, the conservative choice, because it never invents holdings the amendment did not contain.

364 filer-quarters were amended. Median 157 days from original to final version, p90 264, max 389; median absolute book revision 1.09%. A period-keyed pipeline reads the final version at the original date, which is lookahead that leaves no trace – the row looks entirely ordinary.

Data engineering

Every discretionary decision is counted rather than asserted; the rejection ledger is written to disk alongside the holdings and regenerated by `make data-report`.

- **Value units.** SEC Release 34-95148 moved values from thousands to whole dollars for filings made on or after 2023-01-03. A calendar rule fails twice: the cutover keys on the filing date, so 2022Q4 books filed in 2023 report whole dollars, and a tail of filers ignores the change entirely. Scale is inferred per filing from the median implied price; the calendar is only a fallback and is recorded as such.
- **CUSIP.** Mod-10 double-add-double check digit. Leading zeros eaten by a spreadsheet are restored and dropped check digits recomputed, but only when the checksum confirms exactly one candidate; anything ambiguous is dropped rather than guessed. Share classes stay distinct – GOOG and GOOGL have different CUSIPs and must never merge.
- **Filer deduplication.** Union-find over the cover page's otherManagers edges, rebuilt each quarter. A parent files a combination report and the subsidiary files either a notice or its own duplicate table; uncorrected, the same dollars appear twice and a crowding factor reads that as two managers independently agreeing. Name similarity only *proposes* merges for human sign-off in `config/families.yaml`.
- **Instrument class.** Option overlays report notional exposure, not ownership, and PRN lines are face value of debt. Both are removed before any book total is computed, so “largest book in the group” already means largest equity book.
- **Split adjustment.** Each leg of a quarter-on-quarter share change is divided by its *own* quarter's cumulative factor. Applying the current factor to both turns an untouched position across a 4:1 split into a 75% sale.

Hypothesis and factor construction

Concentrated demand from managers with large liquidity footprints has two effects of opposite sign at different horizons. Contemporaneously the buying pushes prices up. Further out, to the extent the buying was pressure rather than information, the push decays. Pure information implies continuation with no reversal; pure pressure implies reversal with no drift. Measuring both horizons is the test, and the interaction with fragility is what separates them: reversal should concentrate where ownership is concentrated and the flow covariance is high.

- **caf** – crowded accumulation: net dollar demand from universe managers over the quarter, divided by quarterly dollar volume. The normalisation makes the unit “days of volume absorbed” rather than raw dollars, which is what determines price impact.
- **consensus** – signed breadth: the fraction of universe holders buying versus selling.
- **own_share** – the share of outstanding held by the universe.
- **fragility** – the Greenwood–Thesmar quadratic form $\theta' \Sigma \theta / W^2$, with a shrunk, PSD-clipped flow covariance. This is the vulnerability-to-forced-selling term.
- **caf_x_fragility** – the interaction, formed on ranks rather than levels so that one extreme observation in either leg cannot manufacture the product.

All factors are winsorised, z-scored cross-sectionally, and residualised on log market cap and log Amihud illiquidity. That last step is not optional here: the signal is built from a liquidity-weighted quantity, so without neutralisation the factor would partly be a restatement of illiquidity.

Backtest

Monthly rebalance with the convention “weights from information at t, return over (t, t+1], costs charged at t”. Two portfolios per factor: an equal-weight Q5 minus Q1 spread, and a z-weighted dollar-neutral book capped at 2% per name. Costs are a 4bp half-spread plus square-root impact using the same law applied to manager footprint, for internal consistency, plus 40bp annual borrow on shorts. Gross and net are both reported.

Between activations the quarterly signal is held flat – no interpolation, no smoothing forward – with a staleness cap so a gap in the filing record produces missing exposure rather than a stale position held forever.

All t-statistics are Newey–West with 6 lags. Monthly returns from a quarterly signal are autocorrelated by construction: the same signal is held for roughly three months, so an OLS t-statistic overstates significance. A deflated Sharpe discounts the best observed result by the number of variants tried.

Results

Simulated data. These numbers describe the generator, not markets.

factor	months	net_ann	vol_ann	sharpe_net	t_nw	maxdd	ic_mean	ic_t_nw	turnover_m	cost_drag_ann
caf_zn	99	+0.70%	+3.58%	0.20	0.51	-10.64%	-0.0008	-0.13	+18.13%	+0.51%
caf_x_fragility_zn	99	+0.88%	+2.64%	0.33	0.71	-6.91%	+0.0068	0.87	+21.13%	+0.55%
consensus_zn	99	+0.38%	+2.07%	0.18	0.43	-7.71%	+0.0019	0.32	+22.49%	+0.52%
own_share_zn	99	+0.50%	+3.17%	0.16	0.42	-6.74%	-0.0011	-0.22	+4.21%	+0.26%
fragility_zn	99	+0.45%	+3.48%	0.13	0.35	-5.79%	+0.0001	0.01	+4.25%	+0.27%

Net of costs. t-statistics Newey–West.

Benchmark attribution

factor	months	raw_ann	alpha_ann	alpha_t_nw	r2	beta_mkt	beta_smb	beta_mom	beta_strev	beta_illiq
consensus_zn	99	+0.38%	+0.58%	0.79	0.06	0.00	-0.16	-0.05	0.04	0.14
caf_zn	99	+0.70%	+0.57%	0.45	0.04	-0.01	-0.08	-0.01	-0.02	0.17
own_share_zn	99	+0.50%	+0.57%	0.47	0.03	0.02	-0.13	0.02	0.02	0.16
caf_x_fragility_zn	99	+0.88%	+0.66%	0.55	0.01	0.00	-0.05	-0.02	-0.02	0.08
fragility_zn	99	+0.45%	+0.12%	0.09	0.04	0.02	-0.16	0.06	-0.02	0.16

Factor returns regressed on market, size, momentum, short-term reversal and illiquidity proxies built from the same price panel. A raw spread that disappears once these are controlled for is one of them wearing a 13F costume. Against vendor data, substitute the

Fama–French and AQR series – the loader accepts them directly.

Horizon IC – the continuation-versus-reversal test

factor	1m	3m	6m	12m
caf_x_fragility_zn	0.0082	0.0182	0.0209	0.0056
caf_zn	-0.0006	-0.0006	-0.007	-0.0158
consensus_zn	0.0023	0.0114	0.0166	-0.0073
fragility_zn	0.0014	0.0038	0.0071	0.0162
own_share_zn	0.0009	0.0026	0.0054	0.0167

The sign pattern across horizons is the hypothesis test, not the Sharpe. A factor that is positive at one month and negative at twelve is behaving like transient pressure; one positive throughout is behaving like information.

Universe stability (last quarters)

period_end	entries	exits
2023-09-30	8	8
2023-12-31	4	4
2024-03-31	3	3
2024-06-30	3	3
2024-09-30	5	5
2024-12-31	6	6

What this does not do

- **13F is long-only, quarterly, US-listed equity.** No shorts, no derivatives exposure beyond the excluded overlays, no cash, no international book. “The manager sold” and “the manager rotated into something 13F cannot see” are indistinguishable here.
- **A quarter-end snapshot is not a trade record.** Round-trips inside the quarter are invisible; window dressing at the boundary is visible and looks like a real position.
- **The overlap measure is symmetric.** It cannot tell a manager who leads a crowd from one who follows it. Sias-style lead-lag on the flow panel is the natural extension and the flow history is already persisted for it.
- **CUSIP reassignment is handled but not exercised.** The security master keys on (cusip, date) and will respect a dated crosswalk, but the fixtures contain no reassignment, so that path is untested end to end. With CRSP it works; that is a claim about the code, not an observation.
- **Fixture results are not evidence.** The generator plants a mechanism and the pipeline recovers it, which tests the plumbing and nothing else.

Next, in priority order

1. **Run against real EDGAR and a survivorship-free price panel.** Everything above is a claim about code until this is done. CRSP is the right source: it handles delisting returns, and forced liquidation on the way to a delisting is one of the mechanisms the factor is trying to capture. A survivorship-biased panel would remove exactly the observations that matter most.
2. **Directional crowding.** Replace symmetric overlap with a lead-lag measure – whose demand precedes whose – which turns centrality from “how connected” into “how upstream”. That is the difference between knowing a stock is crowded and knowing whether the crowd is still arriving or already leaving.
3. **Stress the selection rule.** The 0.40/0.40/0.20 weights and the 25/35 hysteresis band are judgement, not estimates. The honest test is whether results survive a grid over them, reported as a distribution rather than as the best cell – and the deflated Sharpe should be recomputed against the full count of variants examined, not against the handful that reached this memo.

Reproducing

`make setup` && `make test` runs 51 tests with no network. `make demo` generates fixtures and runs the full pipeline offline. Against real data, set `CROWDFLOW_USER_AGENT` to a contact address (SEC returns 403 without one), then `make ingest` and `make run PRICES=... CROSSWALK=...`. A run is fully described by `config/default.yaml` plus a git SHA; the config fingerprint is stamped on every artifact and on this page.